📚 Chapter 6 – Digital Forensics

🎯 Introduction to Digital Forensics

What is Digital Forensics?

🕵️ **Digital Forensics** is the science of identifying, preserving, collecting, analyzing, and presenting digital evidence to investigate cyber incidents, cybercrimes, malware infections, insider threats, and data breaches.

A Digital Forensics Investigator's job is to answer:

✅ What happened?

✅ When did it happen?

✅ Who performed the activity?

✅ How was the attack executed?

✅ What data was affected?

✅ Can it be proven using evidence?

SOC Analyst Perspective

A SOC Analyst monitors attacks.

A Malware Analyst studies malware.

A Threat Hunter searches for hidden threats.

A Digital Forensics Investigator reconstructs the entire attack timeline after the attack occurs.

Therefore, Digital Forensics requires knowledge of:

🦠 Malware Analysis

🌐 Networking

📊 Log Analysis

💻 Windows Internals

🐧 Linux

☁️ Cloud Infrastructure

🛡️ Incident Response

📁 File Systems

🧠 Operating System Architecture

🌍 Real-Life Example

Suppose a company reports:

Windows 11 Machine Infected by Trojan Malware

Management asks:

- Who executed the malware?
- When did it execute?
- Did the user download it?
- Was a USB involved?
- Was company data stolen?

These questions are answered through Digital Forensics.

🎯 What Should Be Checked During a Forensic Investigation?

If a Windows machine is compromised, the investigator examines multiple artifacts.

1️⃣ Login Activity Analysis

Purpose

Determine:

- Who logged in?

- When did they log in?
- Was the login successful?
- Were there brute-force attempts?

Location

Event Viewer
└── Windows Logs
    └── Security

Important Event IDs:

Event ID	Meaning
4624	Successful Login
4625	Failed Login
4634	Logout
4648	Explicit Credentials Used

2️⃣ USB Device Analysis

Why Check USB Activity?

Attackers often:

💾 Copy Data

💾 Deliver Malware

💾 Steal Sensitive Files

through USB devices.

Registry Location

SYSTEM\CurrentControlSet\Enum\USBSTOR

USBSTOR stores history of connected USB devices.

Investigator can determine:

✅ USB Name

✅ Vendor

✅ Serial Number

✅ First Connection Time

✅ Last Connection Time

Example

If a 32 GB SanDisk USB was inserted before confidential files disappeared, it becomes an important lead.

3️⃣ Browser Forensics

Attackers often download malware from websites.

Check:

🌐 Browsing History

🌐 Download History

🌐 Cookies

🌐 Cached Files

🌐 Saved Passwords

Example

Browser History Shows:

crackedphotoshop.com

followed by

Trojan.exe download

This indicates likely malware delivery.

4️⃣ Email Forensics

Many attacks begin through phishing emails.

Check:

📧 Outlook OST Files

📧 PST Files

📧 Attachments

📧 Email Headers

📧 Sender Reputation

Purpose

Determine:

- Who sent the email?
- Was it phishing?
- Did the user open the attachment?

5️⃣ Software Installation Analysis

Determine:

- Which software was installed?
- When was it installed?
- Was it legitimate?

Check:

Program Files

and

Event Viewer → Application Logs

6️⃣ Temporary Files Analysis

Check:

%temp%

Attackers often drop:

🦠 Malware

📜 Scripts

💣 Payloads

into Temp folders.

7️⃣ AppData Analysis

One of the most important forensic locations.

Location:

C:\Users\User\AppData

Contains:

- Browser Artifacts
- Malware Files
- Persistence Scripts
- Temporary Data

Many Trojans and RATs hide here.

8️⃣ Registry Analysis

The Windows Registry stores system configurations and persistence mechanisms.

Investigators check:

Run Keys

RunOnce Keys

Services

Scheduled Tasks

to identify persistence.

🔍 Deleted Data Recovery

Even if files are deleted:

❌ Deleted does not mean destroyed.

Investigators use recovery tools to retrieve:

📁 Deleted Documents

📧 Emails

📸 Images

📜 Logs

Digital Forensics Lifecycle

Step 1 – Identification

Identify:

- Devices
- Accounts
- Logs
- Network Artifacts
- Storage Devices

that may contain evidence.

Step 2 – Preservation

Protect evidence from modification.

This is one of the most important forensic principles.

Write Blocker

What is a Write Blocker?

A Write Blocker is a hardware device that allows:

✅ Read Access

❌ No Write Access

to evidence drives.

Why Use It?

Without a Write Blocker:

Windows may modify:

- Access Time
- Metadata
- Logs

which can invalidate evidence.

Example

Think of a crime scene.

Investigators do not walk over footprints.

Similarly, Digital Forensics avoids modifying original evidence.

Evidence Documentation

Investigators capture:

📸 BIOS Date

📸 BIOS Time

📸 System Time

📸 File Properties

📸 Created Date

📸 Modified Date

📸 Accessed Date

These screenshots help prove integrity of evidence.

Faraday Bag

What is a Faraday Bag?

A Faraday Bag blocks:

📡 Wi-Fi

📡 Bluetooth

📡 Cellular Signals

📡 GPS Signals

Why Use It?

Prevents:

❌ Remote Wipe

❌ Remote Access

❌ Evidence Tampering

Example

A criminal's phone is seized.

If not isolated, the attacker could remotely erase evidence.

The phone is immediately placed inside a Faraday Bag.

Chain of Custody (Very Important)

Definition

Chain of Custody is a legal document that tracks:

- Who collected evidence
- Who handled evidence
- When it was transferred
- Where it was stored

Why Important?

Without Chain of Custody:

🚫 Evidence may be rejected in court.

Example

Evidence ID: 001
Device: Seagate 500GB HDD
Collected By: John Smith
Date: 20-Aug-2025
Time: 10:30 AM

Every transfer is signed and documented.

Forensic Imaging

What is Forensic Imaging?

A forensic image is:

📂 Bit-by-Bit Copy

of a storage device.

Why Not Work on Original Disk?

Because evidence must remain unchanged.

Analysis is always performed on the image copy.

FTK Imager

One of the most widely used forensic imaging tools.

Steps

Create Disk Image

Open:

FTK Imager

↓

Create Disk Image

↓

Physical Drive

↓

Select Drive

↓

Choose Format

E01

↓

Enter:

- Case Number
- Evidence Number
- Investigator Details

↓

Select Storage Location

↓

Set Fragment Size

↓

Start Imaging

What is E01 Format?

E01 is the most common forensic image format.

Benefits:

✅ Compression

✅ Metadata Storage

✅ Integrity Verification

✅ Court Acceptance

Hash Verification

Before and after imaging:

Generate:

🔐 MD5

🔐 SHA1

🔐 SHA256

Why?

If hash values match:

Original Drive Hash
=
Image Hash

then evidence integrity is preserved.

Memory Forensics (RAM Analysis)

Why Analyze RAM?

RAM contains:

🔥 Running Processes

🔥 Active Malware

🔥 Encryption Keys

🔥 Passwords

🔥 Network Connections

Magnet RAM Capture

Used to capture memory from a live machine.

Important instruction:

⚠️ Run from a USB drive.

Do NOT install it on the target system.

Reason:

Installing software can modify evidence.

Network Forensics

Sometimes attackers are still communicating with a server.

Therefore investigators capture:

🌐 Network Traffic

🌐 Packets

🌐 Sessions

🌐 Connections

PCAP Files

Packet captures are stored as:

.pcap

files.

Tools:

- Wireshark
- TCPDump
- NetworkMiner

Types of Digital Forensics

💻 **Computer Forensics**

Analyzing:

- PCs
- Laptops
- Servers

📱 Mobile Forensics

Analyzing:

- Smartphones
- Tablets
- SIM Cards

🌐 Network Forensics

Analyzing:

- Packets
- Network Logs
- Connections

☁️ Cloud Forensics

Analyzing:

- AWS
- Azure
- Google Cloud

📧 Email Forensics

Analyzing:

- Email Headers
- SPF
- DKIM
- DMARC
- Phishing Emails

🦠 Malware Forensics

Analyzing malware behavior and persistence.

Autopsy Tool

What is Autopsy?

Autopsy is a graphical forensic analysis tool.

Official use:

✅ Deleted File Recovery

✅ Browser History Analysis

✅ Registry Analysis

✅ Timeline Creation

✅ Email Analysis

Creating a Case in Autopsy

Step 1

Create New Case

Step 2

Enter:

- Case Name
- Path
- Examiner Name
- Email

Step 3

Add:

- Disk Image
- VM Image

- Local Disk

Step 4

Select Analysis Modules

Step 5

Start Investigation

Slack Space

What is Slack Space?

Suppose:

Cluster Size = 4 KB

File Size = 3 KB

Remaining:

1 KB

is Slack Space.

Deleted data may still exist there.

Investigators recover evidence from Slack Space.

Mobile Forensics

What is Mobile Forensics?

The branch of Digital Forensics focused on smartphones and tablets.

Data Extracted

📞 Call Logs

💬 SMS

📱 WhatsApp Chats

📸 Images

🎥 Videos

📍 GPS History

📧 Emails

👤 Contacts

📁 App Data

Types of Mobile Extraction

Logical Extraction

Visible files only.

Physical Extraction

Complete bit-by-bit copy including deleted data.

File System Extraction

Folder structure and app data.

Cloud Extraction

Data from:

☁️ Google Drive

☁️ iCloud

☁️ WhatsApp Backup

Common Digital Forensics Tools

Tool	Purpose
FTK Imager	Disk Imaging
Autopsy	Disk Analysis
Volatility	RAM Analysis
Wireshark	Network Analysis
Magnet RAM Capture	Memory Capture
EnCase	Enterprise Forensics
X-Ways	Advanced Forensics
Magnet AXIOM	Full Investigation
Cellebrite UFED	Mobile Forensics
PhotoRec	Deleted File Recovery
Scalpel	File Carving

Sample Investigation Flow

Incident Detected
↓
Identify Evidence
↓
Preserve Evidence
↓
Apply Write Blocker
↓
Create Forensic Image
↓
Calculate Hash
↓
Analyze Using Autopsy
↓

Recover Deleted Data
↓
Analyze RAM
↓
Analyze Network Traffic
↓
Create Timeline
↓
Generate Report
↓
Maintain Chain of Custody
↓
Submit Findings

🎤 Important Interview Questions

Q1. What is Digital Forensics?

Digital Forensics is the process of identifying, preserving, collecting, analyzing, and presenting digital evidence for investigation purposes.

Q2. Why is a Write Blocker used?

To prevent modification of original evidence and maintain forensic integrity.

Q3. What is a Forensic Image?

A bit-by-bit copy of a storage device used for investigation without modifying original evidence.

Q4. What is Chain of Custody?

A documented record showing who collected, handled, transferred, and analyzed evidence.

Q5. What is USBSTOR?

A Windows Registry location that stores USB device history.

Q6. Why is RAM Analysis important?

RAM contains live processes, malware, network connections, credentials, and encryption keys.

🎓 Important Certifications (Must Do)

1️⃣ CrowdStrike Certification

Focus:

🛡️ Detection & Response

🛡️ EDR

🛡️ Threat Hunting

🛡️ SOC Operations

Official Website:

[CrowdStrike University]

2️⃣ Qualys Certifications

Recommended Order:

Step 1

✅ Vulnerability Management Foundation

Step 2

✅ VMDR (Vulnerability Management, Detection & Response)

Topics Covered:

🔍 Vulnerability Assessment

🛡️ Detection

🛡️ Risk Management

🛡️ Remediation

Official Website:

[Qualys Training Portal]

📝 **Lecture 6 Quick Revision**

✅ Digital Forensics = Finding evidence after an attack

✅ Check Event Viewer Security Logs

✅ Check USBSTOR Registry

✅ Check Browser History

✅ Check OST/PST Email Files

✅ Check Program Files & Application Logs

✅ Check Temp and AppData Folders

✅ Recover Deleted Files

✅ Use Write Blocker

✅ Store Devices in Faraday Bag

✅ Create Disk Images using FTK Imager

✅ Analyze Images using Autopsy

✅ Capture RAM using Magnet RAM Capture

✅ Analyze PCAP using Wireshark

✅ Maintain Chain of Custody

✅ Verify Evidence using SHA256 Hashes

✅ CrowdStrike & Qualys VMDR certifications are highly valuable for SOC Analyst and Digital Forensics careers. 🚀 🛡️ 🔍